MARIUS EMANUEL HEES

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Erasmus School of Economics, Erasmus University Rotterdam
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Up to date: October 2026

PLACEMENT: Tinbergen Institute
Prof. Eric Bartelsman, placement director
e.j.bartelsman@vu.nl
Christina Månsson, placement assistant
c.mansson@tinbergen.nl

MAIN INTERESTS

I am an economist working on urban, spatial, environmental, and development economics. My research studies how market access, institutional capacity, and beliefs shape economies' resilience to shocks, their path to structural transformation, and their potential for economic growth. Using structural spatial general-equilibrium models, econometric spatial methods with satellite and micro data, and a pre-registered field experiment, I work on urban labor markets and informality, the measurement of local climate shocks, and the beliefs governing climate adaptation, with extensions to housing, insurance, migration, trade, and infrastructure. My agenda takes these increasingly pressing research and policy questions to high-income economies. I am on the 2026–2027 job market.

FIELDS OF CONCENTRATION

PRIMARY
Development Economics
Urban, Spatial and Trade Economics
Environmental and Climate Economics

SECONDARY
Applied Econometrics
Migration and Labor Economics
Quantitative Spatial Economics

DISSERTATION TITLE

"Climate Shocks and Structural Transformation: Satellite Data, Field Experiments and General Equilibrium Models for Resilient Development"

Expected Date of Completion: Spring 2027

REFERENCES

Prof. Frank van Oort (advisor)
Erasmus School of Economics, Erasmus Univ. Rotterdam
vanoort@ese.eur.nl

Dr. Michiel Gerritse (advisor)
Erasmus School of Economics, Erasmus Univ. Rotterdam
gerritse@ese.eur.nl

Prof. Esteban Rossi-Hansberg
Department of Economics, University of Chicago
erossi@uchicago.edu

Prof. Banu Demir Pakel
Department of Economics, University of Oxford
banu.demirpakel@economics.ox.ac.uk

PAPERS

JOB MARKET PAPER

"Segmented Labor Markets and Structural Transformation"

Abstract: Most urban growth this century will occur in developing economies. Whether this urbanization translates into sustained economic development depends on how quickly workers gain access to productive formal employment. I develop a dynamic spatial general equilibrium model with three types of labor markets: rural, urban informal, and urban formal. Migrants enter an informal queue, and formalization depends on informal labor congestion and the economy's capacity to move informal workers into formal employment. I quantify the model for Ethiopia using novel geospatial and microdata. Formalization speed emerges as a key determinant of long-run structural transformation, growth, and welfare. At current capacity, workers wait about 26 years for a formal job and informality persists. Reducing formalization frictions to match rates observed in successfully transforming economies cuts the wait to about 17 years. This raises the urban formal employment share by 26 percentage points and real GDP by 13.1 percent by 2065. Formalization speed shapes the effectiveness of economic policies and national resilience to climate shocks and conflict. Cities become engines of structural transformation not when workers arrive, but when labor markets absorb them into productive employment.

Presented at: North American Meeting of the Urban Economics Association (Chicago, 2026); Dutch Urban Economics and Trade Workshop (2025, 2026).

WORKING PAPERS

"(Mis)measuring Local Weather Shocks: Evidence from Drought and Migration in Ethiopia"
(with Michiel Gerritse and Frank van Oort). Under review.

Abstract: Extreme weather shocks strike locally, yet their effects are typically estimated from exposure aggregated across districts, regions or countries. We show that this aggregation systematically biases impact estimates and propose a simple correction. Using high-resolution rainfall and satellite-derived population data for more than 13,000 Ethiopian settlements (2000–2016), we find that drought reduces population in vulnerable settlements by about 2.5 percent, concentrated among working-age adults and coinciding with falling food production, income and health. Measured at progressively coarser scales, the impact disappears, even at resolutions common in the literature. We decompose the bias into three sources (partial exposure misclassified, local vulnerability averaged away, area weights misrepresenting where people live) and derive a fix: measure each shock at fine scale, weight by local vulnerability, then aggregate. The correction recovers estimates in areas roughly four times larger than those at which the standard binary indicator fails; simulations confirm its performance. Building exposure locally before aggregating is a low-cost change that can substantially improve empirical practice.

Presented at: 6th GCD Workshop on Regional Development and Innovation (A Coruña, 2024); Regional Science Workshop on Remote-Sensed Identification and Networks (Oviedo, 2024); PSE Summer School on Migration (2023); ESE/TI internal seminars.

"Memory, Expectations, and Climate Adaptation"
(with Frank van Oort). Working paper, September 2026. Pre-registered field experiment (AEA RCT Registry).

Abstract: Smallholder farmers face large and growing weather risk, yet take-up of insurance and other protection remains low. The value farmers place on protection depends on the weather they expect. Where records and forecasts are limited, those expectations are built from memories of past weather, and distorted recall can leave demand for protection poorly aligned with the risks farmers actually face. We measure how much experienced weather survives into perceived risk. In a survey experiment with 686 farmers in eastern Uganda, we elicit remembered and expected weather through visual token allocations and map geospatial rainfall and vegetation records onto the same scale. We find that memory is strongly compressed, with about 90 percent of seasonal weather variation lost in the baseline estimate. Yet remembered weather strongly predicts expectations, so this compression carries into beliefs about future risk. By contrast, salient extremes remain vivid. Farmers who recall an extreme are 19 percentage points more likely to expect another and also plan substantially more adaptation. In a randomized experiment, information on past local weather moves expectations toward the record and raises confidence. Accessible historical information can therefore help farmers assess climate risk and inform adaptation decisions in the face of climate extremes.

Presented at: ESE/TI internal seminars.

EDUCATION

	LEVEL	DATE	FIELD
Erasmus University Rotterdam & Tinbergen Institute	PhD	2027 (exp.)	Economics
University of Chicago (Host: E. Rossi-Hansberg)	Visiting PhD	2024	Economics
University of Oxford	MPhil	2022	Economics
Erasmus School of Economics	BSc	2019	Economics
Harvard University	Visiting BSc	2018	Economics

PROFESSIONAL EXPERIENCE

Research Assistant to Dr. Niclas Moneke, University of Oxford 2021
Spatial development project on mobile-phone network expansion and economic development

Research Assistant, ifo Institute for Economic Research, Munich 2020
Project with Prof. Niklas Potrafke on trade networks (Center for Public Finance and Political Economy)

Research Assistant and Consultant, United Nations ESCAP, Bangkok 2019
Cross-national rail connectivity across the Asian continent (Transport Connectivity and Logistics Section)

Research Assistant, University of Tehran 2019
Data collection on local labor market dynamics

TEACHING

Lecturer, Urban Economics (MSc), Erasmus School of Economics 2025
Foundational course of the MSc programme in Urban, Port and Transport Economics. Designed and taught the module on developing-country cities: lecture, R-based problem set and seminar on general-equilibrium analysis of urban transport policy, and assessment

Lecturer, Firm Location Choice and Productivity in Cities (BSc), Erasmus School of Economics 2022–2025
BSc Strategy Economics seminar, four cohorts: weekly session reports, literature surveys, student paper presentations and quizzes; grading and feedback for 100–120 students per cohort

Thesis Supervision, Erasmus School of Economics 2022–2026
BSc theses in Strategy Economics (15) and Urban, Port and Transport Economics (3); MSc theses in Strategy Economics (19) and Urban, Port and Transport Economics (4). Topics include optimal transport-network allocation, carbon pricing and local pollution, energy-crisis resilience of firms, and urban regeneration.

SKILLS

Programming	R, Julia, Python, Stata, MATLAB, Bash, Git, LaTeX
Spatial and remote sensing	ArcGIS, QGIS, Google Earth Engine; large-scale satellite-data pipelines
High-performance computing	Snellius national supercomputer (SURF); parallel and batch workflows
Languages	German (native), English (fluent), French (C1), Dutch (A2)

WORKSHOPS, SEMINARS AND CONFERENCES

North American Meeting of the Urban Economics Association, Chicago	2026
Dutch Urban Economics and Trade Workshop, Groningen	2026
Dutch Urban Economics and Trade Workshop, Rotterdam	2025
6th GCD Workshop on Regional Development and Innovation, A Coruña	2024
Regional Science Workshop on Remote-Sensed Identification and Networks, Oviedo	2024
Summer School on Migration (Giovanni Peri), Paris School of Economics	2023
ESE and Tinbergen Institute internal seminars	2023–2026

FELLOWSHIPS, GRANTS AND AWARDS

Erasmus Research Institute of Management (ERIM) €11,000 2025–2026
Field Experiment in Uganda: PhD Support and Data Collection Grants

Tinbergen Institute, ESE and Erasmus Professor funds €16,000 2025–2026
Field Experiment in Uganda: Internal research funding

Erasmus Trustfonds €5,000 2024
Research visit, University of Chicago

German Academic Scholarship Foundation (Studienstiftung) €40,000 2020–2022
MPhil scholarship

German Academic Scholarship Foundation (Studienstiftung) €45,000 2016–2019
BSc scholarship

OTHER INFORMATION

Citizenship: Germany and France (EU); United Kingdom pre-settled status